

Spatial Statistics Model Evaluation

Predicting Seagrass Habitats

INPUT MODEL FILE

.../SeagrassPrediction_c66cdb/p30/seagrass_model.ssm

ARCGIS PRO VERSION

ArcGIS Pro 3.7.2

MODEL TYPE

Presence-only prediction (without background points)

EXECUTION DATE

Tuesday, October 6, 2026

AREA UNDER CURVE (AUC)

0.9603

OMISSION RATE

0.0459

Model Characteristics

Explanatory Variable Expansions	Original (Linear), Squared (Quadratic), Pairwise interaction (Product), Smoothed step (Hinge)
Number of Knots	10
Study Area Type	Polygon
Spatial Thinning Distance	2 kilometers
Relative Weight (Presence/Background)	100
Link Function	c-log-log
Presence Probability Cutoff	0.5000

Explanatory Training Rasters

Variable Name	Field Type	Description	Unit	Categorical
EBK_DISSO2	Double	Dissolved Oxygen	—	No
EBK_NITRATE	Double	Nitrate Concentration	—	No
EBK_PHOSPHATE	Double	Phosphate Concentration	—	No
EBK_SALINITY	Double	Salinity Levels	None	No
EBK_SILICATE	Double	Silicate Concentration	—	No
EBK_SRTM30	Double	Bathymetry / Elevation	m	No
EBK_TEMP	Double	Temperature	°C	No

Regression Coefficients

Expansion / Interaction Term	Coefficient Value
EBK_PHOSPHATE ²	0.3703
EBK_TEMP ²	0.0000
product(EBK_DISSO2, EBK_TEMP)	0.0021
product(EBK_SALINITY, EBK_SRTM30)	0.0000
product(EBK_SILICATE, EBK_SRTM30)	-0.0030
product(EBK_SILICATE, EBK_TEMP)	-0.0033
product(EBK_SRTM30, EBK_TEMP)	0.0000
hinge(EBK_DISSO2, 7)	0.7869
hinge(EBK_DISSO2, 8)	-2.5903
hinge(EBK_DISSO2, 9)	1.4734
hinge(EBK_DISSO2, 10)	0.0555
hinge(EBK_NITRATE, 4)	0.1913
hinge(EBK_NITRATE, 9)	-0.0214
hinge(EBK_PHOSPHATE, 7)	-4.0232
hinge(EBK_SALINITY, 8)	-0.5496
hinge(EBK_SALINITY, 12)	0.0110
hinge(EBK_SALINITY, 13)	0.0540
hinge(EBK_SRTM30, 7)	0.9154
hinge(EBK_SRTM30, 16)	8.1556
hinge(EBK_TEMP, 7)	0.5758
hinge(EBK_TEMP, 8)	0.1500
hinge(EBK_TEMP, 10)	0.3229
hinge(EBK_TEMP, 11)	1.1438

Explanatory Variable Range Diagnostics

Variable	Minimum Value	Maximum Value
EBK_DISSO2	4.41	7.04
EBK_NITRATE	0.03	12.02
EBK_PHOSPHATE	0.03	1.35
EBK_SALINITY	14.24	36.75
EBK_SILICATE	0.78	22.10
EBK_SRTM30	-202.49	0.82
EBK_TEMP	3.10	26.75

Key Hinge Interval Diagnostics

Representative hinge interval transformations generated for oceanographic variables during model fitting.

Dissolved Oxygen (EBK_DISSO2) Hinge Functions

Knot Order	Variable Function	Lower Bound	Upper Bound	Type
0	hinge(EBK_DISSO2, 0)	4.4135	7.0413	Forward Hinge
2	hinge(EBK_DISSO2, 2)	4.9974	7.0413	Forward Hinge
4	hinge(EBK_DISSO2, 4)	5.5814	7.0413	Forward Hinge
6	hinge(EBK_DISSO2, 6)	6.1653	7.0413	Forward Hinge
8	hinge(EBK_DISSO2, 8)	6.7493	7.0413	Forward Hinge
9	hinge(EBK_DISSO2, 9)	4.4135	4.7055	Reverse Hinge
12	hinge(EBK_DISSO2, 12)	4.4135	5.5814	Reverse Hinge
15	hinge(EBK_DISSO2, 15)	4.4135	6.4573	Reverse Hinge
17	hinge(EBK_DISSO2, 17)	4.4135	7.0413	Reverse Hinge

Sea Surface Temperature (EBK_TEMP) Hinge Functions

Knot Order	Variable Function	Lower Bound	Upper Bound	Type
0	hinge(EBK_TEMP, 0)	3.0977	26.7483	Forward Hinge
2	hinge(EBK_TEMP, 2)	8.3534	26.7483	Forward Hinge
4	hinge(EBK_TEMP, 4)	13.6091	26.7483	Forward Hinge
6	hinge(EBK_TEMP, 6)	18.8648	26.7483	Forward Hinge
8	hinge(EBK_TEMP, 8)	24.1204	26.7483	Forward Hinge
9	hinge(EBK_TEMP, 9)	3.0977	5.7255	Reverse Hinge
12	hinge(EBK_TEMP, 12)	3.0977	13.6091	Reverse Hinge
15	hinge(EBK_TEMP, 15)	3.0977	21.4926	Reverse Hinge
17	hinge(EBK_TEMP, 17)	3.0977	26.7483	Reverse Hinge